Aadya Gupta

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Experience

Dell Technologies *Onsite* — June 2026 - Present
Intern
- Built a multi-agent AI pipeline with specialized agents for feature detection, context aggregation, and audience-aware documentation generation, demonstrating core agentic properties of autonomy, task decomposition, and inter-agent coordination
- Designed a human-in-the-loop feedback mechanism enabling agents to incorporate reviewer signals and detect documentation drift, exploring oversight and continual self-improvement patterns in production agentic systems

Max Planck Institute *Remote* — January 2026 - Present
Research Assistant
- Worked on understanding the internal workings of a language model when subjected to script variations.
- Researched the systemic cause of failure of language models in variable linguistic settings.

Fiery Digital Imaging India Pvt. Ltd. *Hybrid* — May 2025 - July 2025
Machine Learning Intern
- Developed evaluation metrics to assess the accuracy of a language model-based system for automated print-jobs.
- Improved user support by designing a RAG system using internal product manuals for queries.

Publications

- **Gupta A., et al (2026)** ***Understanding How LLMs Process Script Variation : A Behavioral and Representational Analysis*** Under review at EMNLP.
- **Gupta A., et al (2026)** ***A Mechanistic Understanding of Logic Decay In SLMs*** In progress.

Projects

Representing Neural Geometry of Models Processing Code Mixed Data
PyTorch, Sparse Autoencoders
- Probed the internal representational geometry of Qwen3 using activation analysis and dimensionality reduction to map how the model encodes mixed code-language data across layers
- Analyzed cross-modal representational structure in Qwen3 by mapping activation geometry over mixed code-text inputs, contributing to mechanistic understanding of how LLMs internally organize heterogeneous data

Physics-Informed Neural Networks for Glioblastoma Growth Modeling
PyTorch
- Developed a PyTorch PINN to model glioblastoma dynamics, processing BraTS MRI scans with Nibabel and generating synthetic ground-truth via FDM solvers.
- Embedded the Fisher-Kolmogorov equation into the loss function, enabling dynamic parameter discovery of tumor diffusion and proliferation rates.

Education

Manipal University, Jaipur, India
B.Tech in Computer Science and Engineering (Hons.) (AI/ML); GPA: 9.65 — August 2023 - Present

Indian Institute of Technology, Madras, India
Bachelors of Science in Programming and Data Science (Online Dual Degree) — January 2024 - Present
Courses: *DL and GenAI, Machine Learning, Application Development, DSA, RDBMS*

Skills

Programming Languages:	Python, C++, LaTeX, Java, JavaScript, SQL
Frameworks and Libraries:	PyTorch, Hugging Face, Flask
Tools:	Git/GitHub, Jupyter Notebook, Weights & Biases, Linux/Bash

Volunteer Experience

Projects and Research Team Head *ACM-Association for Computing Machinery(MUJ)* — June 2024 - 2025
- Contributed to the field of AI through writing research papers and promoting thought leadership within the chapter.
- Mentored and led a team of 8 talented freshmen and guided them in project building and learning.

Awards and Honors

- Google Girl Hackathon'25 Semi-Finalist
- AWS AI/ML Scholarship Program - worth $4,000 as one of **2,000 global recipients**.
- Dean's List

Courses

- **ARENA** Curriculum
- AI Programming with Python, Nanodegree Program **Udacity**